

Statistics

A Journal of Theoretical and Applied Statistics

ISSN: 0233-1888 (Print) 1029-4910 (Online) Journal homepage: www.tandfonline.com/journals/gsta20

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To cite this article: Alessandro Spelta & Emanuela Raffinetti (2024) Evaluating SAFE AI principles using Wasserstein distance: a comparative study of Machine Learning models, Statistics, 58:6, 1283-1303, DOI: 10.1080/02331888.2024.2419420

To link to this article: https://doi.org/10.1080/02331888.2024.2419420

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Published online: 24 Oct 2024.

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STATISTICS
2024, VOL. 58, NO. 6, 1283–1303
https://doi.org/10.1080/02331888.2024.2419420

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Evaluating SAFE AI principles using Wasserstein distance: a comparative study of Machine Learning models

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ABSTRACT

Modern data-driven Artificial Intelligence (AI), powered by advanced Machine Learning (ML) models, is transforming financial technologies by enhancing financial inclusion, transparency and reducing transaction costs. However, the opaque nature of some complex ML models requires new statistical approaches to manage risks and ensure trustworthiness. In this paper, we present a novel method to evaluate the key principles of trustworthy AI - Sustainability (Robustness), Accuracy, Fairness, and Explainability (SAFE). While Babaei et al. [A Rank Graduation Box for SAFE AI. Expert Syst Appl. 259;2025:125239, 2025] introduced the Rank Graduation Box as a streamlined approach for assessing the principles of trustworthy AI, we extend this work by employing the Wasserstein distance. Our method offers a more nuanced and geometrically oriented comparison of ML models, particularly in contexts where shifts in economic or environmental conditions alter the prediction distributions. We apply this method to compare popular ML models, including Support Vector Machines, Ensemble Trees, K-Nearest Neighbours Linear and Logistic Regression. The proposal is validated using both simulated data and real-world data in the context of financial risk assessment. Our findings demonstrate that the Wasserstein distance offers nuanced and interpretable insights into model behaviour across the SAFE dimensions, making it a valuable tool for model selection and regulatory compliance in AI applications.

ARTICLE HISTORY
Received 20 June 2024
Accepted 10 October 2024

KEYWORDS
Lorenz curves; Optimal Transport; Wasserstein distance; SAFE Artificial Intelligence

2020 MATHEMATICS SUBJECT CLASSIFICATIONS
62R07; 62P20; 68T01

1. Introduction

In the last years, the fast and growing availability of data and computational power has led to the spread of sophisticated Machine Learning (ML) models, together with Artificial Intelligence-based systems. The application of Artificial Intelligence (AI) methods leads to societal and economical benefits resulting into an improvement of user experience and inclusion.

Nevertheless, besides the opportunities derived from its employment, AI can also generate potential risks, such as cyber risks (typically emerging from attacks to the Information and Communication Technology systems) and model risks (caused by the failure of a specific selected model to provide quantitative information).

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Many AI systems, while complex, offer varying degrees of interpretability and control. However, some still present challenges in terms of transparency: data processing and decision-making can occur through intricate mechanisms that may be difficult to fully monitor. This can potentially complicate the management of operational, reputational, and strategic risks, particularly when it comes to identifying and mitigating biases in the system's outputs.

With the aim of avoiding, or at least mitigating, the negative effects induced by an incorrect use of the AI outputs, the European Commission has recently proposed a regulation on Artificial Intelligence, the AI Act, which appears as the first attempt in the world to manage the use of AI by resorting to a set of basic principles addressed to meet the condition of trustworthiness (see [1]). To be trustworthy, AI systems have to be: robust to shock scenarios which may lead to extreme data; accurate in terms of predictions; fair with respect to the sub-groups composing the population; explainable in terms of their predictive output.

In line with the previous requirements, [2–4], among others, illustrated interesting applications of explainable AI methods based on the Shapley's cooperative game theory framework (see [5]).

Recently, further contributions were developed giving rise to new statistical metrics based on the extension of the Lorenz and concordance curves (see [6]). As an example, [7] exploited the Lorenz Zonoid tool, corresponding to the multivariate Gini index (see [8]), to detect the explainability of the predictors in a black-box Machine Learning (ML) model. Differently from Shapley-values [4,5], their proposal is expressed through a normalized scale which measures the percentage of the overall variability explained by each single predictor.

The condition of safety of AI systems implies also the need of evaluating the capability of the highly complex ML models to be accurate. This because the automatic choice of an action is strongly affected by the made predictions. The measures commonly used in literature (RMSE or AUROC) depend on the type of response variable, and none of them can be universally applied, thereby limiting the autonomy of AI applications. This issue becomes more problematic when the response is a vector of variables, of different types (continuous, ordinal, binary), implying difficulties in the choice of the suitable accuracy measure. To fill this gap, [9] introduced a new global measure (called Rank Graduation Accuracy) that can be universally applied, regardless of the nature of the underlying response variable. The Rank Graduation Accuracy (RGA) measure is normalized and based on the comparison between the Lorenz curve of the response variable and the concordance curve obtained ordering the response values in terms of the predicted values. Giudici and Raffinetti [10], further investigated the properties of the RGA metric and showed its equivalence with the AUROC in the classification framework.

Based on these premises, [11] provided a further methodological extension of the Lorenz-Zonoids, called the SAFE approach, with the purpose of assessing also the sustainability (robustness) and fairness principles of AI systems, in addition to those of the accuracy and explainability. The Authors used the term SAFE with a twofold role: with the meaning of reliability of AI but also as the acronym of 'Sustainability', 'Accuracy', 'Fairness', and 'Explainability', i.e., the four principles to be achieved by each model.

For a more complete overview of the main contributions addressed to the assessment of the AI systems, we refer to Professor Paolo Giudici's call for papers in 'Safe Machine Learning' published in this Journal on June 2024 and to the references therein (see [12]). In

response to the same call of papers, [13] investigated the application of Bayesian inference within the SAFE model framework, in order to ensure the accuracy and reliability of the environmental, social, and governance (ESG) conditions within the financial field.

More recently, with the purpose of further improving and facilitating the implementation of the metrics proposed by Giudici and Raffinetti [11], Babaei et al. [14] have introduced a novel framework for the AI risk measurement. Specifically, the approach by Giudici and Raffinetti [11] was re-formalized through the employment of the Concordance curve which appears as a common root for the construction of consistent metrics, all based on the notion of concordance between two cumulative distributions. In this way, a revised set of metrics simple to interpret and extendable to additional AI compliance principles is derived.

Coherently with the contribution by Babaei et al. [14], we propose to investigate the joint dynamics of the scores mentioned above using the Wasserstein distance, derived from the Optimal Transport (OT) theory (see [15,16]). The Wasserstein distance offers several key advantages over previous approaches. It provides a more nuanced comparison by capturing the underlying geometry of distributions, accounting for differences in location and scale that might be overlooked by simpler measures. This is particularly valuable in financial applications, where shifts in economic conditions might alter the location or spread of predictions without fundamentally changing their structure. The Wasserstein distance's ability to compare distributions with different supports or dimensionality also makes our method highly adaptable for future extensions, potentially allowing comparisons between diverse model types or prediction formats. Moreover, its reduced sensitivity to small perturbations or outliers contributes to more stable and robust assessments of model behaviour, a crucial factor when evaluating AI systems for high-stakes applications in finance.

In this paper, we introduce a novel framework that leverages Optimal Transport (OT) theory and the Wasserstein distance to evaluate Machine Learning models against the SAFE AI principles: sustainability (robustness), accuracy, fairness, and explainability. Our approach provides a unified metric for assessing these diverse aspects of model performance, offering a comprehensive view of model trustworthiness as outlined by the AI Act. By applying this framework to various ML models, including Support Vector Machines, Ensemble Trees, K-Nearest Neighbours, Linear and Logistic Regression, we demonstrate its utility in comparing, and selecting appropriate models based on specific application requirements and regulatory guidelines.

The paper is structured as follows. In Section 2, an overview of the existing methodology for the assessment of the safety condition of ML models, together with the illustration of our proposal based on the Waaserstein distance, is provided. In Section 3, a Monte Carlo simulation is led with the purpose of comparing the behaviour of the Wasserstein distance-based metrics with respect to the existing SAFE measures. In Section 4 an application to real data, concerning the context of financial risk assessment, is taken into account. Finally, Section 5 concludes with a discussion about the relevance of the SAFE approach when dealing with AI systems.

2. Methodology

This section is devoted to the illustration of the methodology. Specifically, in Sections 2.1 and 2.2, we describe the mathematical background underlying both the approach

introduced by Babaei et al. [14] and our proposal based on the Optimal Transport Theory and the Wasserstein distance.

2.1. Background

The Rank Graduation Box (RGB), recently proposed by Babaei et al. [14], is based on the notion of concordance between two cumulative distributions and consists of a set of consistent metrics developed for the assessment of the SAFE principles.

Formally, let Y^* and Y^{**} be two statistical variables, each defined on a set of n data points and defined on a continuous, ordinal or binary scale. The RGB metrics are derived through a direct comparison between the cumulative distributions of the Y^* variable values ordered according to three alternative ways: (a) in a non-decreasing sense; (b) in a non-increasing sense; (c) based on the non-decreasing ranks of Y^{**}.

The Rank Graduation Box metrics are expressed through a ratio: the numerator specifies the distance between the cumulative distribution of the Y^* values ordered in a non-increasing sense and that of the Y^* values ordered according to the non-decreasing ranks of Y^{**}; the denominator measures the distance between the cumulative distribution of the Y^* values ordered in a non-increasing sense and that of the Y^* values ordered in a non-decreasing sense.

After some mathematical manipulations, the Rank Graduation $RG\cdot$ index is derived as:

$$RG\cdot = \frac{\sum_{s=1}^{n} sy^*_{r^{**}_s} - \sum_{s=1}^{n} sy^*_{r^*_{n+1-s}}}{\sum_{s=1}^{n} sy^*_{r^*_s} - \sum_{s=1}^{n} sy^*_{r^*_{n+1-s}}}, \tag{1}$$

where, for any $s = 1, \ldots, n$: r^*_s are the non-decreasing ranks of the y^*_s values; r^{**}_s are the non-decreasing ranks of the y^{**}_s values; r^*_{n+1-s} are the non-increasing ranks of the y^*_s values; $\bar{y}^*$ is the mean of the y^*_s values.

From Equation (1), the specific metrics for the measurement of the sustainability, accuracy, fairness and explainability of AI applications are provided by replacing Y^* and Y^{**} with the proper inputs characterizing the four principles:

- to measure sustainability (robustness), the 'Rank Graduation Robustness' (RGR) metric is defined by setting $Y^* = \hat{Y}$ and $Y^{**} = \hat{Y}^p$, where $\hat{Y}$ and $\hat{Y}^p$ denote the predicted values obtained using the non-perturbed data and the predicted values obtained using perturbed data, respectively;
- to measure accuracy, the 'Rank Graduation Accuracy' (RGA) metric is derived by setting $Y^* = Y$ and $Y^{**} = \hat{Y}$, where Y and $\hat{Y}$ denote the observed response values and the corresponding predicted values, respectively;
- to measure fairness, the 'Rank Graduation Fairness' (RGF) metric is derived by setting $Y^* = \hat{Y}$ and $Y^{**} = \hat{Y}^{(-X_g)}$, where, given a set of G group variables (converted into G binary variables), $\hat{Y}$ are the predicted values obtained from the full model (including all the G group variables as predictors) and $\hat{Y}^{(-X_g)}$ are the predicted values obtained from the reduced model (i.e., the model which excludes the g-th group variable under evaluation, with $g = 1, \ldots, G$);

- to measure explainability, the 'Rank Graduation Explainability' (RGE) metric is derived by setting $Y^* = \hat{Y}$ and $Y^{**} = \hat{Y}^{(-X_h)}$, where, given a set of H predictors, $\hat{Y}$ are the predicted values obtained from the full model (which includes all the H predictors) and $\hat{Y}^{(-X_h)}$ are the predicted values obtained from the reduced model (i.e., the model which excludes the h-th predictor under evaluation, with $h = 1, \ldots, H$).

The RG· metric takes values in the close range $[0, 1]$. The more the $RG\cdot$ approaches to value 1, the higher is the robustness, accuracy and fairness of the model. The aforementioned behaviour does not hold for explainability: indeed, the stronger is the effect of a variable X_h on explaining Y, the greater is the divergence between the ranks of the full model predicted values and those associated with the reduced model. It follows that, for coherence with the other metrics, RGE is defined as RGE$= 1-$RG·.

2.2. Proposal

2.2.1. Optimal transport (OT) theory

Optimal Transport (OT) measures the minimum effort needed to reconfigure the probability mass of an input distribution to match a target distribution. Initially formulated as an intractable, non-convex optimization problem, its modern linear programming formulation was introduced by Kantorovich in 1942 (see [17]) and is detailed in Villani's monograph (see [18]). OT primarily addresses the computation of distances between distributions, taking into account the geometry of their supports.

In this work, we focus on the specific case of discrete probability measures, i.e., histograms, in line with [19]. Accordingly, we write the first distribution as $\alpha = \sum_{i=1}^{m} \alpha_i \delta_{I_i}$, where I_i represents the i-th bin interval. This allows us to obtain the point-wise estimated density for a data point x as $\alpha(x) = \sum_{i=1}^{m} \alpha_i \delta_{I_i}(x)$. Similarly, the second distribution is $\beta = \sum_{i=1}^{m} \beta_i \delta_{I_i}$. Both distributions are defined over supports described by the set $\{I_1, \ldots, I_m\}$. The variable δ_{I_i} is the Dirac measure at I_i. Such distributions are fully characterized by the probability mass vectors α_i and β_i defined over the simplex $\Sigma_m \overset{\text{def.}}{=} \{h \in \mathbb{R}_+^m; \sum_i h_i = 1\}$.

We can then express the distance between α and β as:

$$d(\alpha, \beta) = \min_{\pi \in \Pi(\alpha, \beta)} \langle \pi, C \rangle. \tag{2}$$

The transportation plan $\pi \in \Pi(\alpha, \beta)$ determines the joint distribution under the OT metrics, on the product space $\{I_1, \ldots, I_m\} \times \{I_1, \ldots, I_m\}$, where the cost matrix $C \in \mathbb{R}^{m \times m}$ specifies the transportation cost from individual pairs I_i to I_j. Consequently, π is contained in the transportation polytope $\Pi(\alpha, \beta)$, specified as:

$$\Pi(\alpha, \beta) = \{\pi \in \mathbb{R}_+^{m \times m} \mid \pi 1_m = \alpha, \pi^\top 1_m = \beta\}. \tag{3}$$

2.2.2. Wasserstein distance

The Wasserstein distance, also known as the Earth Mover's distance, is a key concept in OT theory. It evaluates the geometric discrepancy between input distributions by determining the minimum 'work' required to transfer all the mass from one distribution to the other. Unlike other metrics such as the Kullback-Leibler divergence or the ℓ_1 and ℓ_2 norms, the Wasserstein metric considers the geometry of the distributions' supports and does not require the distributions to be defined on the same probability space.

By selecting, $C_{i,j} := \|I_i - I_j\|^2$, the OT problem formulated in Equation (2) yields the so-called ℓ_2-Wasserstein distance $d(\cdot,\cdot) = W^2(\cdot,\cdot)$.

The distance $W^2(\alpha,\beta)$ is usually computed by solving the Linear Assignment Problem (LAP) from Equation (2) via specialized algorithms (see [20–22]). However, these approaches are computationally heavy and slow in practice, as pointed out in [23]. A popular alternative is augmenting the LAP objective with an additional entropy parameter, which gives rise to the Sinkhorn operator defined as:

$$\mathrm{OT}(\alpha,\beta) \stackrel{\text{def}}{=} \min_{\pi \in U(\alpha,\beta)} \langle \pi, C\rangle + \gamma\, H(\pi), \tag{4}$$

where $\gamma > 0$ and $H(\pi) \stackrel{\text{def}}{=} \sum_{ij} \pi_{ij} \log \pi_{ij}$ is the negative entropy. Then, $\mathrm{OT}(\alpha,\beta)$ is differentiable and can be solved using Sinkhorn iterations (see e.g., [23]). By defining the Gibbs kernel as $K \stackrel{\text{def.}}{=} \mathrm{e}^{-\frac{C}{\gamma}} \in \mathbb{R}_+^{m\times m}$, and a vector $f = 1_m$, one can find π by using the Sinkhorn iterations:

$$g \leftarrow \frac{\beta}{Kf} \quad \text{and} \quad f \leftarrow \frac{\alpha}{K^\top g}. \tag{5}$$

As shown by Cuturi [23], this scheme converges to a minimizer of Equation (4). Thus the coupling matrix can be obtained as:

$$\pi = \mathrm{diag}(f) K\, \mathrm{diag}(g). \tag{6}$$

More in dept, let y_T^1 and y_T^2, be two 1-dimensional (vector) processes of length T. Moreover, let $\alpha(y_T^1)$ and $\beta(y_T^2)$ be two uniform distributions located on the support $1,\ldots,T$, defined as:

$$\begin{aligned} \alpha(y_T^1) &:= \frac{1}{|T|}\sum_{t\in T} \delta_t, \\ \beta(y_T^2) &:= \frac{1}{|T|}\sum_{t\in T} \delta_t. \end{aligned} \tag{7}$$

In this context, we set the cost between vectors to be half quadratic, namely $C = \frac{1}{2}(\|y_T^1 - y_T^2\|^2)$.

Our method thus provides geometrically meaningful distances to compare discrete distributions, and therefore data which represent the space over which distributions are located. In this way, data can be negative and not summing to one. Moreover, since they enter in the cost matrix, they can be on any possible dimension.

2.2.3. Mathematical foundations of the Wasserstein metric in our framework

To elucidate the workings of the Wasserstein metric in our specific context, we consider the mathematical foundations of our approach. We define our distributions $\alpha(y_T^1)$ and $\beta(y_T^2)$ as uniform distributions located on the support $1,\ldots,T$:

$$\alpha(y_T^1) := \frac{1}{|T|}\sum_{t\in T} \delta_t, \quad \beta(y_T^2) := \frac{1}{|T|}\sum_{t\in T} \delta_t, \tag{8}$$

where δ_t is the Dirac measure at t. Given that both distributions are uniform over the same support, the transportation plan π will be a $T \times T$ matrix with all entries equal to $1/|T|^2$.

This is because each point in the source distribution must be transported equally to all points in the target distribution:

$$\pi_{ij} = \frac{1}{T^2} \quad \forall\, i,j \in \{1,\ldots,T\}. \tag{9}$$

The key difference in our approach lies in the cost matrix C, which we define as:

$$C = \tfrac{1}{2}\|y_T^1 - y_T^2\|^2. \tag{10}$$

This cost matrix captures the squared Euclidean distance between the actual values of y_T^1 and y_T^2, not just their positions in the uniform distribution. Now, we compute the squared 2-Wasserstein distance:

$$\begin{aligned} W^2(\alpha,\beta) &= \min_{\pi\in\Pi(\alpha,\beta)} \langle \pi, C\rangle \\ &= \sum_{i,j} \pi_{ij} C_{ij} \\ &= \frac{1}{|T|^2}\sum_{i,j}\frac{1}{2}(y_i^1 - y_j^2)^2 \\ &= \frac{1}{2|T|^2}\sum_{i,j}(y_i^1 - y_j^2)^2. \end{aligned} \tag{11}$$

This formulation reveals several important properties of our approach:

Scale invariance: the Wasserstein distance is invariant to the scale of T. As $T \to \infty$, the distance converges to:

$$W^2(\alpha,\beta) = \frac{1}{2}\int_0^1\int_0^1 (F_1^{-1}(x) - F_2^{-1}(y))^2\,\mathrm{d}x\,\mathrm{d}y, \tag{12}$$

where F_1^{-1} and F_2^{-1} are the quantile functions of the continuous distributions approximated by y_T^1 and y_T^2.

Sensitivity to value differences: unlike methods that only consider the ranks or positions of values, our approach is sensitive to the magnitude of differences between y^1 and y^2 values.

Quadratic penalization: the squared difference $(y_i^1 - y_j^2)^2$ means that larger discrepancies between y^1 and y^2 values are penalized more heavily than smaller ones.

To illustrate, consider a simple example where $T = 2$, $y^1 = [0,1]$ and $y^2 = [0,2]$. The cost matrix would be:

$$C = \frac{1}{2}\begin{bmatrix} 0 & 1 \\ 1 & 4 \end{bmatrix}. \tag{13}$$

And the Wasserstein distance would be:

$$W^2(\alpha,\beta) = \frac{1}{2\cdot 2^2}(0+1+1+4) = \frac{3}{8} = 0.375. \tag{14}$$

In the context of our SAFE AI evaluation, this formulation allows us to define:

Sustainability (Robustness): we quantify how much model predictions change under data perturbations, capturing both frequency and magnitude of changes:

$$W_{SUST} = W^2(\hat{y}, \hat{y}_{\text{perturbed}}). \tag{15}$$

Accuracy: we measure the discrepancy between predictions and actual values, sensitive to the size of prediction errors:

$$W_{ACC} = W^2(y, \hat{y}). \tag{16}$$

Fairness: we assess how model behaviour changes across different subgroups, detecting both subtle and large disparities:

$$W_{FAIR} = \max_{g,g'} W^2(\hat{y}_g, \hat{y}_{g'}). \tag{17}$$

Explainability: we quantify the impact of removing variables on model predictions, capturing the magnitude of changes in the prediction distribution:

$$W_{EXP} = W^2(\hat{y}, \hat{y}_{-h}), \tag{18}$$

where $\hat{y}_{-h}$ are predictions with variable x_h removed. This mathematical foundation underscores the power and flexibility of our Wasserstein distance-based approach in providing a unified, interpretable framework for evaluating key aspects of AI trustworthiness.

2.2.4. Application to machine learning models

The Wasserstein distance provides a versatile framework for evaluating various aspects of ML models. For Support Vector Machines (SVM), we can use the Wasserstein distance to measure the change in the decision boundary under perturbations, providing insights into model robustness. In Ensemble Trees, such as Random Forests or Gradient Boosting Machines, the Wasserstein distance can quantify the diversity of predictions across individual trees, offering a measure of model uncertainty and stability. For K-Nearest Neighbours (KNN), the Wasserstein distance can assess the impact of changes in the feature space on the model's predictions, providing a measure of local sensitivity. In Linear and Logistic Regression, we can use the Wasserstein distance to evaluate the change in predictions under feature perturbations or across different subgroups, offering insights into model fairness and stability. By applying the Wasserstein distance to these diverse model types, we can provide a unified and interpretable metric for assessing their performance across the SAFE dimensions, facilitating meaningful comparisons and informed model selection.

3. Wasserstein metrics versus alternative competitors

To contextualize our Wasserstein distance-based method within the existing literature, we conducted, by means of simulated data, a comparative analysis with the RGB-based approach proposed by Babaei et al. [14].

3.1. Simulation conditions

In order to set the simulation study, we referred to the experiments A and C reported in the paper by Babaei et al. [14]. Specifically, according to experiment A, 1000 five-dimensional

Table 1. Correlation matrix S_1 (Source: Babaei et al. [14]).

	Y	X_1	X_2	X_3	X_4
Y	1	0.6	0.4	0.3	0.1
X_1		1	0.2	0.7	0.3
X_2			1	0.05	0.1
X_3				1	0.5
X_4					1

samples (Y, X_1, X_2, X_3, X_4) with sample size equal to 100 observations were drawn from a Normal distribution with a mean vector $\mu = \{150, 20, 72, 5, 200\}$ and a correlation matrix S_1 specified as in Table 1.

To replicate experiment C, 1000 samples from a non-Normal distribution characterized by the same correlation matrix with skewness and kurtosis parameters equal to $\nu = 3$ and $\kappa = 61$ were also drawn.

Moreover, perturbed data were generated by replacing in all the four predictors the values lower than the 15% percentile and greater than the 85% percentile with values sampled from uniform distributions $U(-6, -4)$ and $U(15, 22)$, respectively.

Finally, for assessing fairness, the predictor X_1 was binarised assigning label '1' to the values greater than the mean and label '0' in the opposite case. By doing so, variable X_1 is treated as a group variable.

Four different ML models were run on data from experiments A and C: a Linear Regression model; an Ensemble Tree model; a K-Nearest Neighbour (KNN) model; a Support Vector Machine (SVM) model. Linear Regression model predicts the value of a variable as a function of one or more independent variables. It belongs to the class of models which are explainable by design. The K-Nearest Neighbour, instead, predicts the class of a data point based on the majority class of its k closest neighbours in the feature space. In contrast, Ensemble Tree is a ML technique that handles non-linearities and interactions without needing a specified regression model. It uses bootstrap aggregation of multiple regression trees to mitigate overfitting, combining predictions from numerous trees for more accurate forecasting. Finally, the Support Vector Machine is a supervised learning algorithm used for classification that finds the optimal hyperplane which maximizes the margin between different classes in the feature space.

We remark that all the models were fit on the same training and test sets, including the 80% and the remaining 20% of the observations, respectively.

With the aim of properly show how to assess the robustness, accuracy, explainability and fairness of ML models, the Wasserstein distance metrics were computed and compared with the ones proposed by Babaei et al. [14].

3.2. Simulation results

In this subsection we investigate the contribution of the Wasserstein distance and RGB metrics in assessing the key aspects of AI safety. The results of the metrics are expressed by the corresponding mean values computed along the 1000 drawn samples.

To further validate the empirical findings, the JT test is applied (see [24]). The JT test compares q metrics ν_r (with $r = 1, \ldots, q$). Under the null hypothesis, there is no difference among the expected values of the q metrics. Whereas, under the alternative hypothesis of

monotonic ordering, the expected values differ. Note that, in terms of accuracy, robustness and fairness, the models are ranked from the lowest Wasserstein distance value, onwards; and from the highest RG· value, downwards. It follows that the hypotheses to be specified for the RG· metrics take the form of $H_0 : \nu_1 = \cdots = \nu_q$ vs $H_1 : \nu_1 > \cdots > \nu_q$, while for the Wasserstein distance metrics the alternative hypothesis is defined as $H_1 : \nu_1 < \cdots < \nu_q$.

Explainability is assessed by interpreting the RGE and the Wasserstein distance from the highest value, downwards, as both the metrics move in the same direction. Indeed, for a specific predictor, a high RGE value means that the model is strongly affected by its deletion. This happens also in the case of the Wasserstein distance metric, as dropping the predictor from the model implies that the distance between the predictions from the full model and those from the model without the predictor of interest increases. Thus, the JT test alternative hypothesis associated with the Wasserstein distance metric has to be set as follows: $H_1 : \nu_1 > \cdots > \nu_q$.

3.2.1. Accuracy

The predictive accuracy of the models is determined through the RGA measure and the Wasserstein distance metric W_{ACC}.

In formulas, the two metrics are computed in each sample as:

$$\mathrm{RGA}^{SVM}(y, \hat{y}^{SVM}) \quad \mathrm{RGA}^{ENS}(y, \hat{y}^{ENS})$$
$$\mathrm{RGA}^{KNN}(y, \hat{y}^{KKN}) \quad \mathrm{RGA}^{LIN}(y, \hat{y}^{LIN})$$

and

$$\mathrm{W}^{SVM}_{ACC}(y, \hat{y}^{SVM}) \quad \mathrm{W}^{ENS}_{ACC}(y, \hat{y}^{ENS})$$
$$\mathrm{W}^{KKN}_{ACC}(y, \hat{y}^{KNN}) \quad \mathrm{W}^{LIN}_{ACC}(y, \hat{y}^{LIN}),$$

where superscripts *SVM*, *ENS*, *KNN* and *LIN* refer to the Support Support Vector Machine, Ensemble Tree, K-Nearest Neighbour, Linear Regression models; y and $\hat{y}$ are the observed response values and the predictions computed on the test set.

Tables 2 and 3 report the values of the two metrics (RGA and W_{ACC}), jointly with the ranking of the models based on their capability to provide accurate prediction in the case of normally and non-normally distributed data.

From Table 2, it arises that both the approaches detect Linear Regression model as the first ranked model and the SVM as the model which provides an intermediate contribution to prediction.

The model rankings induced by the RGA and the W_{ACC} metrics are confirmed by the JT test ($p-$values < 0.01), meaning that Linear Regression actually appears as the most accurate model.

Table 2. RGA and W_{ACC} metrics computed between target data and model predictions (normal data).

Ranked model	RGA	Ranked model	W_{ACC}
LINEAR REG	0.8297	LINEAR REG	0.4278
ENSEMBLE	0.7911	KNN	0.5067
SVM	0.7865	SVM	0.5073
KNN	0.7728	ENSEMBLE	0.5514

Table 3. RGA and W_{ACC} metrics computed between target data and model predictions (non-normal data).

Ranked model	RGA	Ranked model	W_{ACC}
LINEAR REG	0.8263	LINEAR REG	0.4429
ENSEMBLE	0.8001	KNN	0.5261
SVM	0.7902	ENSEMBLE	0.5517
KNN	0.7831	SVM	0.5519

Similar conclusions can be derived from Table 3: also in this scenario, the RGA and the W_{ACC} metrics detect the Linear Regression as the most accurate model. The JT test further validate the model rankings provided by both the metrics ($p-$values < 0.01). However, the findings from non-normal data show that based on the W_{ACC} metric, the model characterized by an intermediate contribution to the predictive accuracy is the Ensemble Tree instead of the K-Nearest Neighbour model.

3.2.2. Sustainability

The robustness dimension is assessed in each sample through the RGR and W_{SUST} metrics, whose formulas are reported below:

$$\mathrm{RGR}^{SVM}(\hat{y}, \hat{y^p}^{SVM}) \quad \mathrm{RGR}^{ENS}(\hat{y}, \hat{y^p}^{ENS})$$
$$\mathrm{RGR}^{KNN}(\hat{y}, \hat{y^p}^{KKN}) \quad \mathrm{RGR}^{LIN}(\hat{y}, \hat{y^p}^{LIN})$$

and

$$\mathrm{W}^{SVM}_{SUST}(\hat{y}, \hat{y^p}^{SVM}) \quad \mathrm{W}^{ENS}_{SUST}(\hat{y}, \hat{y^p}^{ENS})$$
$$\mathrm{W}^{KKN}_{SUST}(\hat{y}, \hat{y^p}^{KKN}) \quad \mathrm{W}^{LIN}_{SUST}(\hat{y}, \hat{y^p}^{LIN}),$$

where $\hat{y}$ are the model predictions derived from data without perturbations and $\hat{y^p}$ are the model predictions derived from data with perturbations.

Results on robustness are displayed in Table 4 (normal data) and Table 5 (non-normal data).

From Tables 4 and 5, it results that the RGR and W_{SUST} metrics take values equal to 1 and 0, respectively, identifying the SVM model as perfectly robust to data perturbations. It is worth noting that the model rankings in Tables 4 and 5 fully overlap, apart from the case of the RGR metric with non-normally distributed data, where it arises that Linear Regression has a better performance than the Ensemble Tree and K-Nearest Neighbour models, as also supported by the JT test $p-$values ($p-$values < 0.01).

Table 4. RGR and W_{SUST} metrics computed between model predictions derived from data without perturbations and model predictions derived from data with perturbations (normal data).

Ranked model	RGR	Ranked model	W_{SUST}
SVM	1	SVM	0
ENSEMBLE	0.9440	ENSEMBLE	0.1789
KNN	0.8773	KNN	0.2381
LINEAR REG	0.6570	LINEAR REG	0.6610

Table 5. RGR and W_{SUST} metrics computed between model predictions derived from data without perturbations and model predictions derived from data with perturbations (non-normal data).

Ranked model	RGR	Ranked model	W_{SUST}
SVM	1	SVM	0
LINEAR REG	0.9737	ENSEMBLE	0.1769
ENSEMBLE	0.9618	KNN	0.2936
KNN	0.8464	LINEAR REG	0.6452

3.2.3. Fairness

We move to the assessment of the fairness dimension, whose findings, included in Tables 6 and 7, are obtained by computing in each sample the RGF and the Wasserstein distance metric W_{FAIR}. In formulas:

$$\begin{aligned} &\mathrm{RGF}^{SVM}_{-X_g}(\hat{y}^{SVM}, \hat{y}^{SVM}_{-X_g}) \quad \mathrm{RGF}^{ENS}_{-X_g}(\hat{y}^{ENS}, \hat{y}^{ENS}_{-X_g}) \\ &\mathrm{RGF}^{KKN}_{-X_g}(\hat{y}^{KNN}, \hat{y}^{KKN}_{-X_g}) \quad \mathrm{RGF}^{LIN}_{-X_g}(\hat{y}^{LIN}, \hat{y}^{LIN}_{-X_g}), \end{aligned}$$

and

$$\begin{aligned} &\mathrm{W}^{SVM}_{FAIR_{-X_g}}(\hat{y}^{SVM}, \hat{y}^{SVM}_{-X_g}) \quad \mathrm{W}^{ENS}_{FAIR_{-X_g}}(\hat{y}^{ENS}, \hat{y}^{ENS}_{-X_g}) \\ &\mathrm{W}^{KKN}_{FAIR_{-X_g}}(\hat{y}^{KNN}, \hat{y}^{KKN}_{-X_g}) \quad \mathrm{W}^{LIN}_{FAIR_{-X_g}}(\hat{y}^{LIN}, \hat{y}^{LIN}_{-X_g}), \end{aligned}$$

where $\hat{y}$ and $\hat{y}_{-X_g}$ (with $g = 1, \ldots, G$) represent the predicted values of the full model (fitted by including all the predictors) and the predicted values from the model fitted by dropping the group variable X_g.

From Table 6, it results that the RGF and W_{FAIR} metrics identify the K-Nearest Neighbour and the Support Vector Machine models as the most and least fair models, respectively. While in the case of the RGF, Ensemble Tree is more fair than Linear Regression, the

Table 6. RGF and W_{FAIR} metrics computed between the full model predictions and the model predictions obtained by dropping the binarised X_1 predictor (normal data).

Ranked model	RGF	Ranked model	W_{FAIR}
KKN	0.9450	KKN	0.0677
ENSEMBLE	0.9130	LINEAR REG	0.1273
LINEAR REG	0.7260	ENSEMBLE	0.1346
SVM	0.6814	SVM	0.2154

Table 7. RGF and W_{FAIR} metrics computed between the full model predictions and the model predictions obtained by dropping the binarised X_1 predictor (non-normal data).

Ranked model	RGF	Ranked model	W_{FAIR}
KKN	0.9405	KKN	0.0519
ENSEMBLE	0.8843	ENSEMBLE	0.1447
SVM	0.8026	SVM	0.2385
LINEAR REG	0.7771	LINEAR REG	0.3149

two models are reversed based on the W_{FAIR} metric. These conclusions are also confirmed by the JT test ($p-$values < 0.01).

Similar findings arise from Table 7: the only difference with respect to the case of normally distributed data, which is in turn supported by the JT test ($p-$values < 0.01), is that the RGF and W_{FAIR} metrics give rise to the same model rankings.

3.2.4. Explainability

Explainability is mainly addressed to the detection of the explanatory variable resulting as the most powerful in affecting the response explanation. To evaluate it, the RGE and the Wasserstein distance W_{EXP} metrics are computed by resorting to the following formulas:

$$\begin{aligned} &RGE^{SVM}_{-X_h}(\hat{y}^{SVM}, \hat{y}^{SVM}_{-X_h}) \quad RGE^{ENS}_{-X_h}(\hat{y}^{ENS}, \hat{y}^{ENS}_{-X_h}) \\ &RGE^{KKN}_{-X_h}(\hat{y}^{KNN}, \hat{y}^{KKN}_{-X_h}) \quad RGE^{LIN}_{-X_h}(\hat{y}^{LIN}, \hat{y}^{LIN}_{-X_h}), \end{aligned}$$

and

$$\begin{aligned} &W^{SVM}_{EXP_{-X_h}}(\hat{y}^{SVM}, \hat{y}^{SVM}_{-X_h}) \quad W^{ENS}_{EXP_{-X_h}}(\hat{y}^{ENS}, \hat{y}^{ENS}_{-X_h}) \\ &W^{KKN}_{EXP_{-X_h}}(\hat{y}^{KNN}, \hat{y}^{KKN}_{-X_h}) \quad W^{LIN}_{EXP_{-X_h}}(\hat{y}^{LIN}, \hat{y}^{LIN}_{-X_h}), \end{aligned}$$

where $\hat{y}$ and $\hat{y}_{-X_h}$ (with $h = 1, \ldots, H$) represent the predicted values of the full model (fitted by including all the predictors) and the predicted values from the model fitted by dropping the h-th predictor X_h.

Tables 8 and 9 display the RGE and W_{EXP} in the case of normally and non-normally distributed data.

Based on Table 8, it follows that, in general, one of the predictors which contributes the most to the explanation of the response is X_1 (both in terms of *RGE* and W_{EXP}). Also the X_2 predictor appears as a relevant variable.

Similarly, from Table 9, it results that X_1 and X_2 are crucial variables for the explanation of the response. However, note that also variable X_4 appears as one of the most important

Table 8. RGE and W_{EXP} metrics computed between the full model predictions and the model predictions obtained by dropping the X_h predictor (normal data).

	Variable X_1		Variable X_2		Variable X_3		Variable X_4	
Model	RGE	W_{EXP}	RGE	W_{EXP}	RGE	W_{EXP}	RGE	W_{EXP}
SVM	0.1057	0.0636	0.0175	0.0993	0.0446	0.0701	0.0175	0.0614
ENSEMBLE	0.1126	0.1470	0.0691	0.1357	0.0358	0.1067	0.0422	0.1257
KNN	0.1924	0.1826	0.1826	0.1958	0.1235	0.1761	0.1315	0.1723
LINEAR REG	0.0278	0.0465	0.0237	0.0985	0.2639	0.0383	0.0032	0.0342

Table 9. RGE and W_{EXP} metrics computed between the full model predictions and the model predictions obtained by dropping the X_h predictor (non-normal data).

	Variable X_1		Variable X_2		Variable X_3		Variable X_4	
Model	RGE	W_{EXP}	RGE	W_{EXP}	RGE	W_{EXP}	RGE	W_{EXP}
SVM	0.0969	0.0785	0.0638	0.0658	0.0111	0.0481	0.1390	0.0577
ENSEMBLE	0.1079	0.1350	0.0772	0.1279	0.0209	0.0857	0.0229	0.0912
KNN	0.1938	0.1738	0.1856	0.1648	0.1105	0.1252	0.0072	0.1421
LINEAR REG	0.0950	0.0537	0.0638	0.0646	0.0072	0.0112	0.1105	0.0161

in terms of RGE when considering Support Vector Machine and Linear Regression models. The $p-$values < 0.01 of the JT test further validate all these conclusions.

Our analysis reveals that while both methods identify similar trends in model performance, the Wasserstein distance-based approach shows increased sensitivity to subtle differences in distribution shapes. Specifically, in terms of fairness evaluation, our approach provides a more granular view of the magnitude of the differences. For explainability, the SAFE approach defines easily interpretable percentage contributions for each variable, while our method offers a more nuanced view of how variable exclusion affects the overall prediction distribution. This suggests that the two methods could be complementary, with SAFE providing an accessible overview and our approach offering deeper insights into distribution changes. These comparisons highlight the unique strengths of our Wasserstein distance-based method, particularly in capturing subtle distribution changes and giving a unified framework across all evaluation dimensions.

4. Application

In this section we aim at illustrating how our proposal works on data referred to the field of financial risk assessment, where the response to be predicted is expressed through a binary scale.

While in the continuous scenario, the RGB and Wasserstein distance-based approaches share the same inputs, i.e., the predictions derived from the model, in the classification framework the RGB method necessarily requires the use of the predicted scores, associated with the true category of the response variable to be predicted. It follows that the Wasserstein distance appears as an extension of the RGB-based approach: indeed, it directly allows the employment of the predicted classifications (expressed through the labels '0' and '1') in place of the predicted scores.

To further emphasize this point, the analysis on data is led by only focusing on the Wasserstein distance computed by means of the predicted classifications.

4.1. Data description

We refer to a subset of the data supplied by Modefinance, a European Credit Assessment Institution (ECAI) that specializes in credit scoring for P2P (Peer-to-Peer) platforms which concern with SME (Small and Medium Enterprises) commercial lending.

Data report the official financial ratios derived from the balance-sheets of 3453 small and medium enterprises. Specifically, the focus is on ten variables: Total Assets/Total Liabilities; Current ratio; Leverage ratio; ROE (Return on Equity); Quick ratio; ROI (Return on investments); Asset turnover; Debt ratio; ROCE (Return on capital); NACE (Nomenclature of Economic Activities) group, i.e., the company's macro-sector of activity specified through three levels: Agriculture; Industry; Services.

In addition, information about the status of the company are also provided through the variable Default ($0 =$ active, $1 =$ defaulted).

An exhaustive description of similar data can be found in [25].

We aim at estimating the default status of each company, using the available explanatory data.

For comparison purposes, we focus on the same models illustrated in the simulation study. We remark that, as dealing with the binary classification problem, the Linear Regression is now replaced by the Logistic Regression model.

4.2. Results

We first explore the sustainability of the aforementioned models. Namely, we compare (in terms of Wasserstein distances), the outputs provided by the models against those obtained by perturbing the explanatory variables.

In particular, we randomly select the 10% of the available data and we add a Gaussian noise to each of them, with mean equal to the standard deviation of the respective variable. We repeat this process 100 times and, at each iteration, we compute the Wassertein distance (W_{SUST}).

Figure 1 shows the histograms of the Wasserstein distances of each model to measure the robustness against data manipulation.

In particular, from Figure 1, it is worth to notice that while the predictions of the K-Nearest Neighbour model do not show any deviation when data are perturbed, those of the Support Vector Machine report the lowest distance as much of the predictions are concentrated in the small values bin of the distribution.

To compare the accuracy of each model we simply compute the Wasserstein (W_{ACC}) distance between the model's prediction and the target variable.

Figure 2 reports the model accuracy in terms of the Wasserstein distance of the predictions from the target variable. As Figure 2 suggests apart from the Ensemble Tree, all the models perform equally well, with the logistic predictions having the lowest distance from the target data.

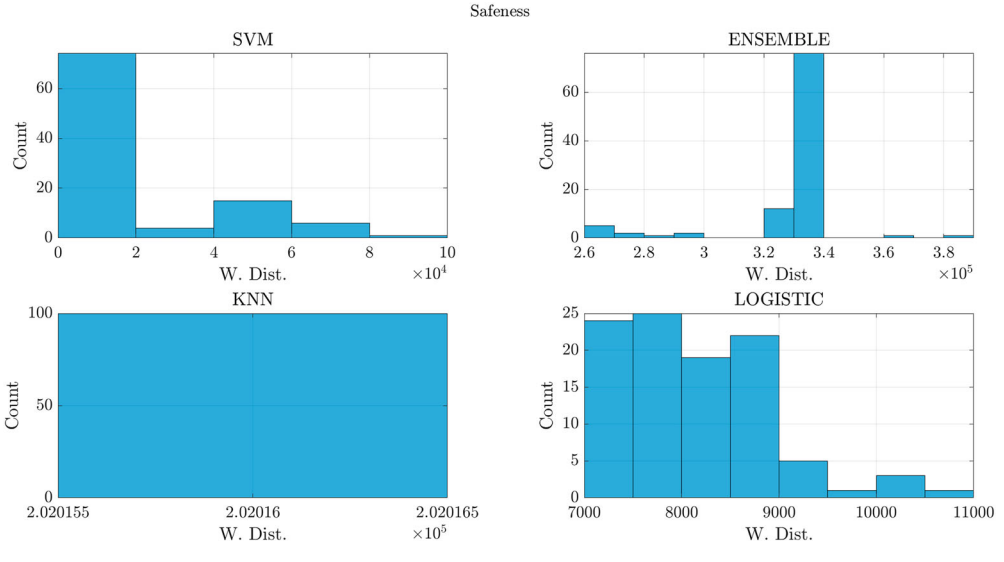


Figure 1. Distribution of Wasserstein distances. The figure reports the histograms of the Wasserstein distances of each ML model as proxy of the robustness of the model against data perturbation.

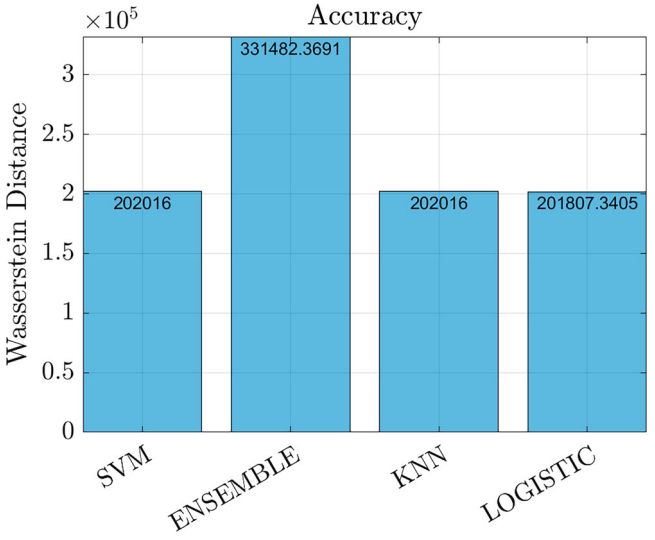


Figure 2. Wasserstein distances between model predictions and target data. The figure reports the Wasserstein distances between model output of each ML model and the target variable, as proxy of the accuracy of the model.

Table 10. Wasserstein distances between model predictions with and without companies with a particular NACE code.

	Agriculture	Industry	Service sector
SVM	0	0	0
ENSEMBLE	2.9092×10^5	13084	2.424×10^5
KNN	0	0	0
LOGISTIC	7157.8	1088	4775.3

Note: The table reports the Wasserstein distances between model outputs before and after having dropped companies that belong to a specific NACE category.

To inspect the models' behaviour in terms of fairness, we exploit the categorical variable which describes the NACE category each company belongs to. Basically, we compare each model outcome before and after having dropped a specific NACE category.

Table 10 shows the fairness of each model and the Wasserstein distance (W_{FAIR}) of its outcomes before and after having dropped companies that belong to a specific NACE category (where $c = 1, 2, 3$, with '1', '2', and '3' corresponding to Agriculture, Industry and Services sectors).[1]

As Table 10 suggests, Support Vector Machine and K-Nearest Neighbour models are not affected by the exclusion of companies of any category, while the Ensemble Tree and the Logistic Regression, beside being affected, display the lowest distance in the case of suppression of companies belonging to the NACE-2 category.

Finally, we are interested in comparing models while accounting for explainability. That is, we ask ourselves whether some variables are more important than other in forming the models outcomes. In order to cope with this issue, we compute the Wasserstein distance (W_{EXP}) of the predictions generated by each model with and without a specific variable.

Table 11 reports the results concerning the explainability of each model, computed as the Wasserstein distances between model outputs before and after having dropped a specific explanatory variable.

As for the fairness, we observe that the Support Vector Machine and K-Nearest Neighbour models are not affected by the exclusion of explanatory variables, while the Ensemble Tree and a Logistic Regression suffer when variables are removed. In particular, the Ensemble Tree is the most affected model, displaying the highest distance.

4.3. Concluding comments

This analysis provides an in-depth analysis of various ML models in terms of their robustness, accuracy, fairness, and explainability.

The K-Nearest Neighbour model demonstrated the highest robustness to data perturbations, with minimal deviations observed. The Support Vector Machine also showed considerable robustness, with its predictions concentrated in the lower values of the distance distribution. In terms of accuracy, the Logistic Regression model outperformed others. Apart from the Ensemble Tree, which showed higher distances, all models generally performed well. The Support Vector Machine and K-Nearest Neighbour models were unaffected by the exclusion of any category. The same holds for the explainability where these models do not heavily depend on any single explanatory variable.

Overall, the findings highlight that while the Support Vector Machine and K-Nearest Neighbour models excel in robustness and fairness, the Logistic Regression model stands out in terms of accuracy. The Ensemble Tree, despite its sensitivity, provides valuable insights but requires careful handling due to its dependence on specific variables. This comprehensive evaluation underscores the importance of selecting appropriate models based on the specific requirements of robustness, accuracy, fairness, and explainability in AI applications.

5. Discussion

The prioritization of robustness, accuracy, fairness, and explainability in AI systems is highly context-dependent and should be carefully considered for each application. In high-stakes decision-making scenarios, such as credit risk assessment for large corporate loans, explainability might take precedence. Here, the ability to provide clear, understandable reasons for a model's predictions is crucial for regulatory compliance and building trust with stakeholders. Conversely, in high-frequency trading applications, accuracy and robustness might be prioritized.

The rapid pace of transactions and the potential for significant financial losses make it essential that models perform consistently and accurately under varying market conditions. However, this doesn't negate the importance of explainability in these contexts, particularly for regulatory oversight and risk management.

For consumer-facing applications like personal loan approvals or insurance pricing, fairness often becomes a top priority. Ensuring that models don't discriminate based on protected characteristics is not only an ethical imperative but often a legal requirement. However, this must be balanced with accuracy to ensure the financial viability of these products.

Table 11. Wasserstein distances between model predictions with and without a particular explanatory variable.

	ROE	Leverage	Asset	Current R.	Quick R.	ROI	Turnover	Debt R.	ROCE	NACE
SVM	0	0	0	0	0	0	0	0	0	0
ENSEMBLE	331,482	331,482	331,482	331,482	331,482	331,482	332,878	331,482	331,482	331,482
KNN	0	0	0	0	0	0	0	0	0	0
LOGISTIC	12,772	9773	11,793	11,768	12,111	12,284	12,101	11,021	12,521	10,561

Note: The table reports the Wasserstein distances between model outputs before and after having dropped a specific explanatory variable.

Robustness becomes particularly crucial in applications subject to adversarial attacks or significant data shifts. For instance, in fraud detection systems, the ability to maintain performance in the face of new, previously unseen fraud patterns is vital.

It's important to note that these dimensions are often interrelated. For example, improving explainability can often lead to insights that enhance fairness and robustness. Similarly, a more robust model might naturally achieve better accuracy across diverse scenarios.

Ultimately, the ideal balance of these dimensions should be determined through careful consideration of the specific use cases, regulatory requirements, ethical implications and potential risks associated with model failures in each dimension.

In evaluating our method against the SAFE approach proposed by Babaei et al. [14], it becomes evident that both methodologies offer distinct advantages in different contexts. Our use of the Wasserstein distance provides a more geometrically nuanced evaluation of Machine Learning models by accounting for differences in both location and scale of distributions. This feature is particularly beneficial in financial applications, where the ability to detect shifts in economic conditions is critical. The flexibility of the Wasserstein distance also allows for comparisons across various types of models and data formats, including categorical and continuous variables. Moreover, the robustness of this metric ensures stability in the face of small perturbations or outliers, which is crucial when dealing with high-stakes financial data. As a unified framework for evaluating the SAFE principles (sustainability, accuracy, fairness, and explainability), our approach offers a comprehensive tool for assessing AI model performance.

However, the computational complexity of our method, especially in large datasets, can be a drawback. The calculations required for the Wasserstein distance are more intensive than those in Babaei et al.'s RGB, which may limit its practical application in time-sensitive or resource-constrained environments. Furthermore, while our approach provides deeper insights, its reliance on Optimal Transport theory might make the results less immediately interpretable for practitioners who are unfamiliar with the underlying mathematics.

In contrast, Babaei et al.'s RGB method offers a more streamlined and computationally efficient set of metrics for evaluating AI models according to the SAFE principles. The RGB approach is grounded in the use of concordance curves, which are simpler and easier to interpret, making it accessible to a broader audience. The internal consistency of their metrics ensures that all four key dimensions of AI safety are evaluated within a unified statistical framework, and its efficiency makes it especially suitable for large-scale applications or real-time monitoring. However, the simplicity of the RGB method comes at the cost of depth. Indeed, if on the one hand the approach proposed by Babaei et al. [14] is excellent for general assessments, on the other hand it may overlook more subtle distributional changes, particularly in complex financial scenarios where the shifts in prediction patterns are not easily captured by rank-based methods alone. Additionally, the RGB approach may be less effective in handling highly irregular or complex distributions, where the geometric precision of the Wasserstein distance becomes more valuable.

These considerations suggest that while our Wasserstein distance-based approach is more robust and adaptable in complex, high-risk environments, the RGB method offers a more practical solution for simpler, routine applications, or when computational resources are limited.

In addition, the Wasserstein distance metrics allow to deal with the classification framework not only by means of the predicted scores, as done by the RGB-based method, but also by means of the predicted classifications as inputs.

In conclusion, by integrating insights from both methodologies, we offer a more comprehensive understanding of model behaviour across various applied contexts.

Note

1. We compare the models' predictions only for the companies that remain after excluding those belonging to a specific NACE category.

Acknowledgments

We also thank the Editor and the two anonymous reviewers for their stimulating comments and remarks which allowed to improve the paper.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work has been partially supported by the Italian Ministry of University and Research (MUR) through the project 'A geo-localized data framework for managing climate risks and designing policies to support sustainable investments' (No. 20229CWYXC) within the PRIN 2022 programme, funded by the European Union – Next Generation EU.

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